

AFSANA KHAN

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EDUCATION

Ph.D. <i>Dissertation: “Unlocking value of Data with Vertical Federated Learning”</i>	Nov. 2021 – Jun. 2026 Maastricht University
M.Sc in Computer Science (Specialization: Distributed Systems) <i>Thesis: “Synthetic Sensor Data Generation for Authentic IoT Device Emulation”</i>	Sep. 2019 – Aug. 2021 University of Tartu
B.Sc in Computer Science and Engineering <i>Thesis: “A New Algorithmic Approach to Finding Minimum Spanning Tree”</i>	Jan. 2014 – Jan. 2018 Bangladesh University of Professionals

EXPERIENCE

Researcher <i>Department of Advanced Computing Sciences, Maastricht University</i> - Worked on machine health monitoring within the DigiMach Interreg project, focusing on tool wear prediction from industrial sensor and image data. - Developed deep learning models for high-frequency sensor streams and image data, including <i>CNN</i> architecture such as <i>ResNet</i> and investigated multimodal fusion for tool-wear prediction. - Designed and implemented a federated learning approach to support collaborative model training across metal industry partners without sharing raw operational data.	Nov. 2025 – Apr. 2026
Ph.D. Researcher <i>Department of Advanced Computing Sciences, Maastricht University</i> - Conducted a structured literature review of Vertical Federated Learning (VFL); synthesized the state of the art, mapped key challenges, and outlined future research directions. - Designed and implemented a single-round, communication-efficient VFL method using autoencoder based compression and secure aggregation, which preserved privacy while yielding a federated model that outperformed individual local models. - Developed a novel secure pre-training participant-selection method for VFL using secure multiparty computation (SMPC) and forward selection strategy to identify high-value parties; outperformed baseline approaches in accuracy and efficiency. - Developed a cooperative game theory based incentive mechanism for VFL that provably ensures fair participant rewards for participating in collaboration. - Developed a hybrid federated learning algorithm for data distributed across horizontal and vertical partitions, and applied it to suspicious financial transaction detection use case using AMLSim and financial data provided by SWIFT. - Provided teaching support for the Masters level courses Data Fusion and Data Mining, which included developing lab materials, delivering some lectures, and grading student assignments.	Nov. 2021 – Oct. 2025
Graduate Research Assistant <i>Mobile and Cloud Laboratory, Institute of Computer Science, University of Tartu</i> Contributed as a Masters student to EU Horizon project <i>RADON</i> and Estonian research project <i>CIISIM</i> by engineering and deploying key technical components, including scalable cloud infrastructure on OpenStack and a proof-of-concept system for synthetic IoT data generation.	Feb. 2020 – Jun. 2021
Lecturer <i>Daffodil International University</i> Fulfilled a range of academic responsibilities, including delivering undergraduate lectures, supervising student projects, and organizing university workshops	May. 2018 – Aug. 2019

TECHNICAL SKILLS

Programming Languages: Python (Proficient), Java (Basic)

Data Wrangling & Analytics: pandas, NumPy, Data Preprocessing, Feature Engineering, Data Visualization

Machine Learning & Modeling: Supervised Learning, Deep Learning, Multimodal Learning, Federated Learning

Libraries & Frameworks: PyTorch, scikit-learn, Flower (federated learning framework)

Tools & Infrastructure: Git, Docker, AWS (S3, Lambda), OpenStack

PUBLICATIONS

- **A. Khan**, M. Stallmann, M. Pietrasik, C. Kouzinopoulos, and A. Wilbik, “Federated Learning for Distributed CNC Tool Wear Prediction,” *Proceedings of the 28th International Symposium on Stabilization, Safety, and Security of Distributed Systems*, 2026. [Accepted]
- **A. Khan**, M. ten Thij, G. Tang, and A. Wilbik, “HybridFL: A Federated Learning Approach for Financial Crime Detection,”. *Proceedings of the 4th International Conference on Federated Learning Technologies and Applications*, 2026. [Accepted]
- **A. Khan**, M. ten Thij, G. Tang, F. Thuijsman, and A. Wilbik, “Fair Incentive Allocation in Vertical Federated Learning Using Nucleolus,” in *Learning-Driven Game Theory for AI*, Elsevier, 2026.
- **A. Khan**, M. ten Thij, and A. Wilbik, “Vertical Federated Learning: A Structured Literature Review,” *Knowledge and Information Systems*, Springer, Apr 2025.
- **A. Khan**, M. ten Thij, G. Tang, and A. Wilbik, “VFL-RPS: Relevant Participant Selection in Vertical Federated Learning,” at International Joint Conference on Neural Networks, 2025. [Oral Presentation].
- **A. Khan**, M. ten Thij, F. Thuijsman, and A. Wilbik, “Using the Nucleolus for Incentive Allocation in Vertical Federated Learning,” *Proceedings of the 2024 2nd IEEE International Conference on Federated Learning Technologies and Applications*, 2024 [Best Paper Award].
- **A. Khan**, M. ten Thij, and A. Wilbik, “Communication-Efficient Vertical Federated Learning,” *Algorithms*, Volume 15, Issue 8, Article 273, 2022.
- **A. Khan**, A. Anzum, J. Sarker, S. Rahman, and J. Rahman, “An IoT-Based Intelligent Fire Evacuation System,” *Proceedings of the 21st International Conference of Computer and Information Technology*, 2018.
- **A. Khan**, A. Anzum Aesha, and J. Sarker, “A New Algorithmic Approach to Finding Minimum Spanning Tree” *Proceedings of the 1st IEEE International Conference on Electrical Engineering and ICT*, 2018.

HONOURS AND AWARDS

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| • Best Paper Award at FLTA, 2024 | 2024 |
| • Achievement Stipend granted by University of Tartu | 2021 |
| • European Union-funded IT Academy Specialization Stipend | 2020, 2021 |
| • Tuition waiver scholarship granted by University of Tartu | 2019 |
| • Bangladesh Board Scholarship for Higher Secondary School Certificate Examination | 2013 |

REFERENCES

Prof. Anna Wilbik

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Dr. Marijn ten Thij

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